Isteyak

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Technical Skills

Programming Languages: Python, C++, C, Rust, Java, JavaScript, TypeScript, SQL, Bash
Web Development (MERN & Full-Stack): MongoDB, Express, React, Node.js, Next.js, Tailwind CSS, Framer Motion, Zustand, Supabase, PostgreSQL, GraphQL, REST APIs, FastAPI, HTML/CSS, Cloudflare Workers, Three.js
Machine Learning & Computer Vision: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLOv8, YOLO11, MediaPipe, RetinaFace, MTCNN, Autoencoders, U-Net
Geospatial Deep Learning: Satellite Imagery Processing, GDAL, Rasterio, QGIS, ArcGIS, MapLibre GL JS
LLM & AI APIs: Cohere, Gemini API, Anthropic API, ElevenLabs, Hugging Face Transformers
Data Analysis & Visualization: pandas, NumPy, SQL, TIBCO Spotfire, Power BI, Excel, Streamlit, Web Scraping (bs4), Kaggle, Google Colab
Edge AI, Embedded & Systems: CUDA, TensorRT, ONNX, ESP32 (ESP-IDF), WebAssembly, Linux, WSL
Robotics & Simulation: NVIDIA Isaac Sim, RMPflow Motion Planning, Action Graph, Model Context Protocol (MCP), Franka FR3
DevOps & Tools: Docker, Git, Kubernetes, GitHub Actions, CI/CD, MLflow, GCP, AWS EC2, uv
Currently Upskilling In: ROS 2, Gazebo, URDF, MultiSet VPS, 6-DoF Camera Localization

Experience

GeoVision Lab Feb 2025-Mar 2026
Researcher *University of Windsor*

- Developed a patching system for creating splits of any satellite image and tested using QGIS render tech.
- Built a web scraping system to extract structured data from database websites, and trained a Siamese Attention-UNet (TensorFlow/Keras) on geospatial image pairs to detect visual similarity for change detection.
- Optimized model training under constrained GPU resources (NVIDIA GTX 1080) by tuning batch size, input resolution, and memory growth settings to avoid OOM errors, then scaled the full training run to an NVIDIA RTX 4080 (Ada Lovelace), cutting training time to under an hour on heavy datasets.

University of Windsor Oct 2024-Dec 2024
Research Assistant *Windsor, ON*

- Researched and evaluated frontal face detection techniques by integrating YOLOv8n with MTCNN and YOLO11 with RetinaFace, achieving improved detecting accuracy and frame-level face counts in ML experiments.
- Conducted performance benchmarking and model comparison to assess inference speed, scalability, and detection reliability across the YOLOv8n/MTCNN and YOLO11/RetinaFace pipelines.
- Collaborated under the supervision of Dr. Ziad Kobti, Former President of the Canadian AI Association, contributing to innovative approaches and automating development workflows for experiments.

Projects

AI-Driven Robot Simulation on Cloud GPU (NVIDIA Isaac Sim + MCP) Oct 2026

- Deployed NVIDIA Isaac Sim 5.1 headless on a cloud RTX 4090 (Vast.ai) and connected it to Claude Code through the open-source isaacsim-mcp-server, building and running robots and physics scenes from plain-English prompts.
- Built a Franka FR3 pick-and-place task with RMPflow motion planning and an Action Graph state machine that reads object positions at runtime, verifying frame by frame that the cube landed within 1.2 cm of target in 3 logged runs.
- Contributed a headless browser live view to the open-source project (PR #50): a --live-view launcher flag with MJPEG streaming, camera controls, unit tests, and docs, verified in a running Isaac Sim and against CI checks.

MLOps Pipeline & CUDA Image Filter Lab Jul 2026

- Built mlops-base, a config-driven MLOps pipeline in Python (scikit-learn) for training, evaluating, and predicting with a classification model, exposed via CLI entry points (mlops-train, mlops-eval, mlops-predict).
- Exposed the trained model through a FastAPI inference API (/health, /predict, launched via mlops-api) with YAML-based configuration, optional MLflow experiment tracking, and pytest smoke tests for reliability.
- Developed a CuPy-based CUDA/GPU-accelerated image-filtering benchmark (blur and edge detection) with an interactive Next.js 15, React 19, and TypeScript frontend for live demonstration.

Tour2Toronto | *CUTC: Transform Hackathon* Aug 2026

- Built a live map of real-time downtown Toronto events in a 3-hour hackathon sprint using Next.js (App Router, TypeScript), MapLibre GL JS, and Zustand, with time-aware pins that pulse for live events and fade for ended ones.
- Aggregated and deduplicated events across the Ticketmaster Discovery API, City of Toronto open-data festivals feed, and OpenStreetMap Nominatim geocoding using a haversine-distance dedup algorithm (˜150m threshold) so each event appears once on the map, plus an AI-powered chat widget for event discovery.

AI-Powered 3D Virtual Classroom | *HackTheNorth 2025* Sep 2025

- Contributed to building an immersive 3D classroom in Three.js with interactive areas for collaborative learning.
- Used Cohere's Command A Vision Model to turn handwritten notes into LaTeX study guides in 3 seconds.

ESP32 Wireless Media Bridge (Edge File Transfer Middleware) Feb 2026

- Built an ESP32-based wireless middleware that enables direct photo and video transfer from a smartphone to a laptop using Wi-Fi AP mode and an embedded HTTP server, with no internet connection required.
- Wrote ESP-IDF firmware for networking, memory constraints, and reliable data streaming on limited hardware.

Ultimate Posture Watcher | *UOttahacks 2026* Jan 2026

- Built a privacy-first, real-time posture coach using webcam input, MediaPipe Pose (WASM), and ergonomic angle.
- Designed configurable alert system (beeps + cooldown logic) and session analytics (posture time, worst angles).

Face Detection and Count using Autoencoder and YOLO ML models Sep 2024

- Used autoencoder to embed each image into a single vector for feature extraction and clustering.
- Applied KNN algorithm to cluster similar faces and the accuracy of the heuristic is recorded for evaluation.

Global Electricity Resources Availability Web App | *WinHacks 2024 Hackathon* Feb 2024

- Assisted in API integration using React, Geocodify, and Electricity Map for real-time data visualization.
- Collaborated in a team of 4 to develop a resource solution, securing 3rd place in the competition.

Education

University of Windsor Windsor, ON

Bachelor of Computer Science *Jan 2023-Dec 2024*

- Transferred Credits from Independent University, Bangladesh (Jan 2021 -- Aug 2022)
- Earned Dean's Awards for Fall 2021 and Winter 2022.